TARUN SINGH

PhD Scholar | Department of Civil Engineering | IIT Roorkee

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RESEARCH PROFILE

PhD researcher working on constitutive modelling of rate-dependent materials, with emphasis on finite-deformation frameworks, rheological connections, evolving natural configurations, and thermodynamically consistent modelling of viscoelastic solids.

EDUCATION

PhD, Civil Engineering | 2023–Present
Indian Institute of Technology Roorkee
Research: Constitutive Modelling of Rate-Dependent Materials | Supervisor: Dr. Sandipan Paul

M.Tech., Structural Engineering | 2020–2022
National Institute of Technology Jalandhar
Thesis: Carbonation Resistance of Concrete Made with Iron Slag and Recycled Concrete Aggregate

B.Tech., Civil Engineering | 2015–2019
Rewa Engineering College, Rewa, Madhya Pradesh
Project: Development of Porous Asphalt Pavement

RESEARCH EXPERIENCE

- PhD research on constitutive modelling of rate-dependent materials and finite-deformation viscoelasticity.
- Development of constitutive frameworks using rheological connections and multiple/evolving natural configurations.
- Formulation and numerical integration of nonlinear constitutive equations, with parameter identification using experimental response data.
- M.Tech. research on durability and performance of concrete incorporating iron slag and recycled concrete aggregates.

SELECTED PUBLICATIONS

Singh, T. and Paul, S. (2026). “A deformation mode-specific constitutive framework for finite viscoelasticity based on scalar, conjugate stress/strain pairs: Applications to characterization of soft materials.” *Journal of the Mechanics and Physics of Solids*, Article 106895. DOI: 10.1016/j.jmps.2026.106895

Singh, T. and Paul, S. (2026). “Constitutive modeling of Kelvin–Voigt type viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations.” *International Journal of Solids and Structures*, 341, Article 114336. DOI: 10.1016/j.ijsolstr.2026.114336

Singh, T. and Paul, S. (2025). “On the extension of the concept of rheological connections to a finite deformation framework using multiple natural configurations.” *International Journal of Engineering Science*, 212, Article 104267. DOI: 10.1016/j.ijengsci.2025.104267

Singh, N., Singh, A., Ankur, N., Kumar, P., Kumar, M., and Singh, T. (2022). “Reviewing the properties of recycled concrete aggregates and iron slag in concrete.” *Journal of Building Engineering*, 60, 105150.

Singh, N., Ankur, N., Singh, T., Kumar, M., Singh, A., and Kumar, P. (2022). “Non-destructive evaluation of iron slag concrete containing recycled concrete aggregates.” *Indian Concrete Journal*, 96(11).

Singh, N., Singh, T., Kumar, M., Singh, A., and Kumar, P. (2022). “Investigating the fresh state performance of concrete containing iron slag and recycled concrete aggregates.” *Materials Today: Proceedings*, 65, 1467–1477.

CONFERENCE PRESENTATIONS

Singh, T. and Paul, S. (2026). “Constitutive modeling of viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations.” *IUTAM Symposium: Rate Dependent Mechanics and Multiscale Modelling of Materials*, IIT Delhi.

Singh, T. and Paul, S. (2026). “Thermodynamically consistent constitutive model for viscoelastic solids under finite deformation using evolving natural configurations.” *7th Indian Conference on Applied Mechanics*, IIT Kanpur.

Singh, T. and Paul, S. (2025). “Challenges in extending rheological connections to a finite deformation theory.” *14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025)*, IIT Roorkee.

RESEARCH INTERESTS

Constitutive modelling of rate-dependent materials • Finite-deformation continuum mechanics • Finite-strain viscoelasticity and viscoplasticity • Rheological models and connections • Evolving and multiple natural configurations • Computational mechanics and finite-element implementation • Sustainable construction materials

TECHNICAL SKILLS

Scientific computing: MATLAB, Python
Finite-element and constitutive implementation: Abaqus, UMAT, Fortran, FEniCSx
Numerical methods: Numerical integration of constitutive equations, parameter identification and model calibration

ACADEMIC PROFILES

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